

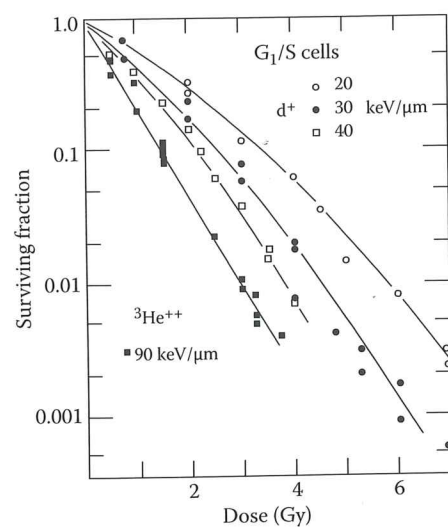


FIGURE 5.13 Survival curves for V79 cells exposed to different charged particles of different LET (Birds et al. 1980).

remains relatively insensitive to the changes of the fraction size. This implies that the ratio of the TD of low- and high-LET radiation required to produce the same effect increases with increasing the number of fractions or, equivalently, with decreasing the dose per fraction (Withers et al. 1982).

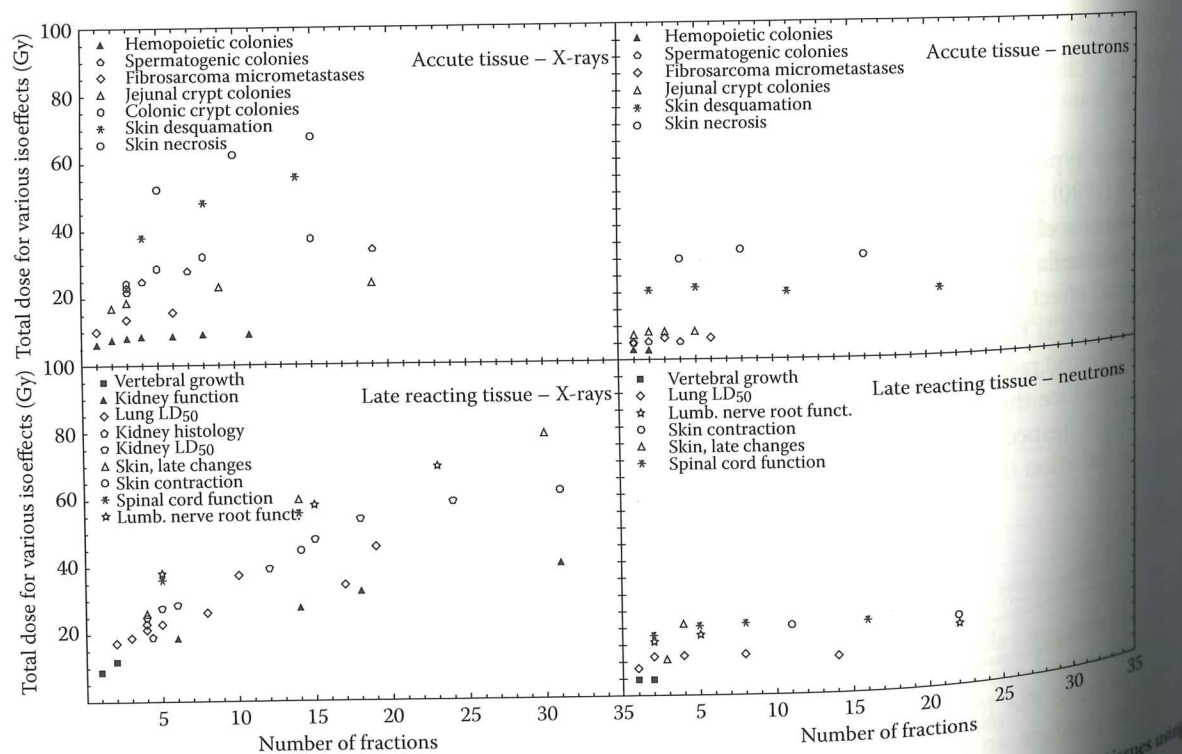


FIGURE 5.14 Total isoeffective doses versus the number of fractions for different early (top) and late (bottom) reacting tissues using low- (X-rays, left) and high-LET (neutrons, right) radiations.

Based on this context, the difference in biological effectiveness of low- and high-LET radiations (RBE) can be formulated as the ratio of the TD required with both types of radiation to produce the same effect, $RBE = \frac{TD_{low-LET}}{TD_{high-LET}}$. This idea is represented in

Figure 5.15a, where the solid line corresponds to the total isoeffective dose for low-LET radiations and the dashed line represents the isoeffective line for high-LET radiations. To understand the shape of these lines, we need to remember that, at very large doses per fraction (i.e., low number of fractions), lethality is mainly produced by the accumulation of sublethal damage, which is represented by the quadratic term in the linear quadratic formulation. This implies that the TD required when delivered in a small number of large fractions (high dose) is lower than in more fractionated regimens in order to produce the same effect (e.g., same number of lethal events). Figure 5.15 also shows how the TD ratio (RBE) increases with an increasing number of fractions, and vice versa. This implies that RBE must have a maximum (RBE_{max}) corresponding to the ratio of the asymptotic values of low- and high-LET TDs at an infinite number of fractions and a minimum (RBE_{min}) corresponding to the ratio of TDs

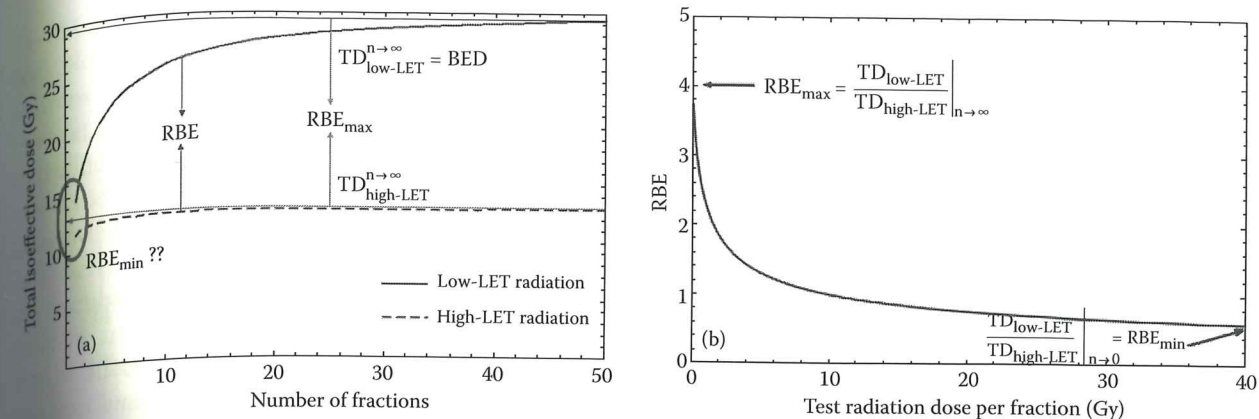


FIGURE 5.15 Variation of RBE with (a) total isoeffective dose and (b) dose per fraction.

obtained from the intersection of the solid and dashed lines with the y-axes, as shown in Figure 5.15b. Due to the asymptotic nature of RBE_{min} , as it corresponds to zero number of fractions, it is virtually impossible to know if it is equal to one (which would imply that both lines in Figure 5.15a intersect the y-axes at the same point) or if it has a value different than one.

The values of RBE_{max} and RBE_{min} can be easily obtained from the linear-quadratic (LQ) formulation as follows. If the selected end point is the production of a given number (N) of lethal events at both high- and low-LET, then

$$N_{low-LET} = N_{high-LET} \Rightarrow \alpha_L d_L + \beta_L d_L^2 = \alpha_H d_H + \beta_H d_H^2 \quad (5.7)$$

At very low doses, the quadratic terms of Equation 5.7 are negligible compared to the linear components; hence, Equation 5.7 can be reduced to

$$\alpha_L d_L = \alpha_H d_H \Rightarrow RBE_{max} = \frac{d_L}{d_H} \Big|_{d \rightarrow 0 Gy} = \frac{\alpha_H}{\alpha_L} \quad (5.8)$$

In the opposite end, at very high doses, the linear terms can be neglected, and Equation 5.7 becomes

$$\beta_L d_L^2 = \beta_H d_H^2 \Rightarrow RBE_{min} = \frac{d_L}{d_H} \Big|_{d \rightarrow \infty Gy} = \sqrt{\frac{\beta_H}{\beta_L}} \quad (5.9)$$

Although there are substantial data for protons (Paganetti et al. 2002) and carbon ions (Ando and

Kase 2009), indicating that $\alpha_L \neq \alpha_H$ for both animal and human *in vitro* and *in vivo* experiments, no such body of data exist to substantiate that $\beta_L \neq \beta_H$. Recent efforts to prove the existence of $RBE_{min} \neq 1$ for neutrons (Jones 2010) and for carbon ions (Carabe-Fernandez et al. 2010) seem to suggest better prediction of the TD required in hypofractionation treatments with these particles when it is assumed that $RBE_{min} \neq 1$ than in the opposite case. The analysis for protons is inconclusive.

Although the use of the LQ formulation has been justified for doses up to 18 Gy (Brenner 2008), it is practically impossible to extrapolate the LQ formulation to dose levels of the order of those required to calculate RBE_{min} . However, it seems more feasible to make such extrapolation to zero number of fractions on fractionated data such as the one shown in Figure 5.16 (left). However, in this case, to obtain reliable values of RBE_{min} that could have an impact on the RBE relevant to hypofractionated regimens, a fractionation data set covering from a very low to a very large number of fractions is required. Very limited data of this type exist for high-LET radiations.

Figure 5.16 shows the difference in the predicted value of RBE at large doses per fraction when RBE_{min} is assumed to be equal to (gray curve) and different (black curve) from 1. In particular, changes have been reported (Carabe-Fernandez et al. 2010) on proton and carbon ion RBE between 7% and 9% at large doses per fraction (see Figure 5.17) when it is considered that $RBE_{min} \neq 1$. It is also suggested that these differences may be larger for tissues with low (α/β) than tissues with high (α/β). This may have important implications in hypofractionated treatments for hepatocellular cancer treated with 14 Gy/fraction carbon ions (Tsujii et al. 2007) and stage I lung cancer treated with proton